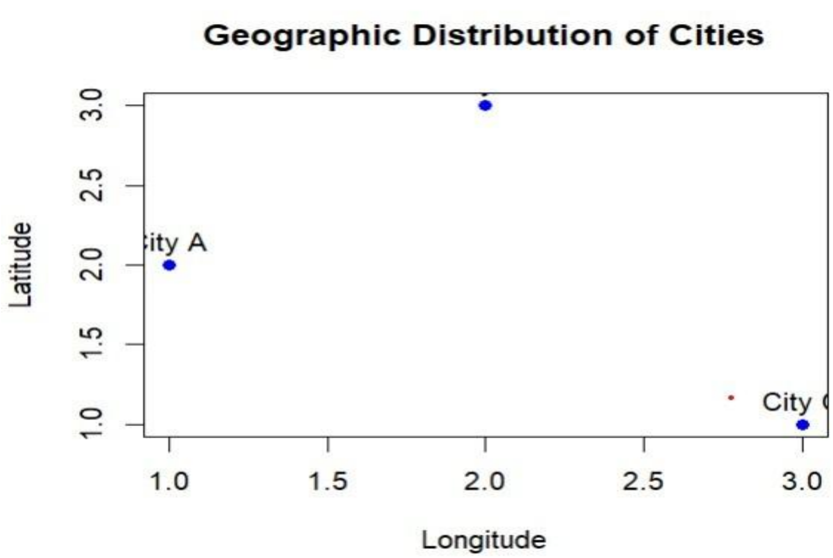
Lab Experiments 13 – 17

SET 13 – Q1 Create a Map Chart

```
# Install package (Run only once) install.packages("leaflet")
library(leaflet)
# Dataset geo <-
data.frame(
  City = c("City A", "City B", "City C"),
  Latitude = c(13.0827, 12.9716, 17.3850),
  Longitude = c(80.2707, 77.5946, 78.4867),
  Population = c(500000,700000,600000)
)
# Map Chart leaflet(geo) %>%
addTiles() %>%   addMarkers(
lng = ~Longitude,     lat =
~Latitude,     popup = ~paste(
"<b>", City, "</b><br>",
      "Population:", Population
    )
  )
```



SET 13 – Q2 Scatter Plot (Population vs Temperature) install.packages("ggplot2")

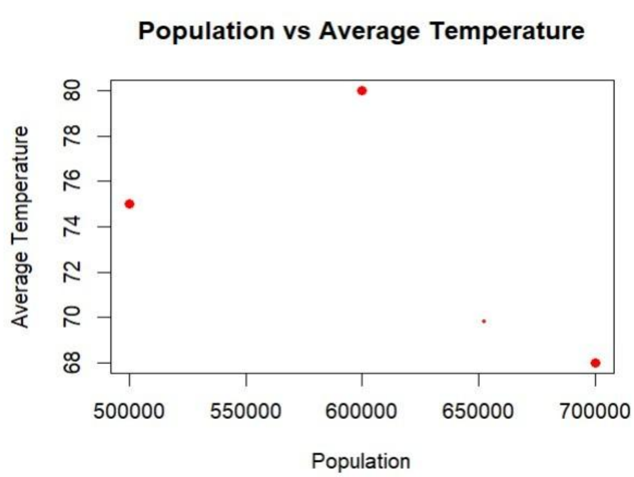
library(ggplot2)

geo <- data.frame(
 City=c("City A","City B","City C"),
 Population=c(500000,700000,600000),
 AvgTemperature=c(75,68,80)
)

ggplot(geo, aes(x=AvgTemperature, y=Population, label=City))+
geom_point(size=5,color="blue")+

geom_text(vjust=-1)+

labs(title="Population vs Average Temperature", x="Average Temperature", y="Population")



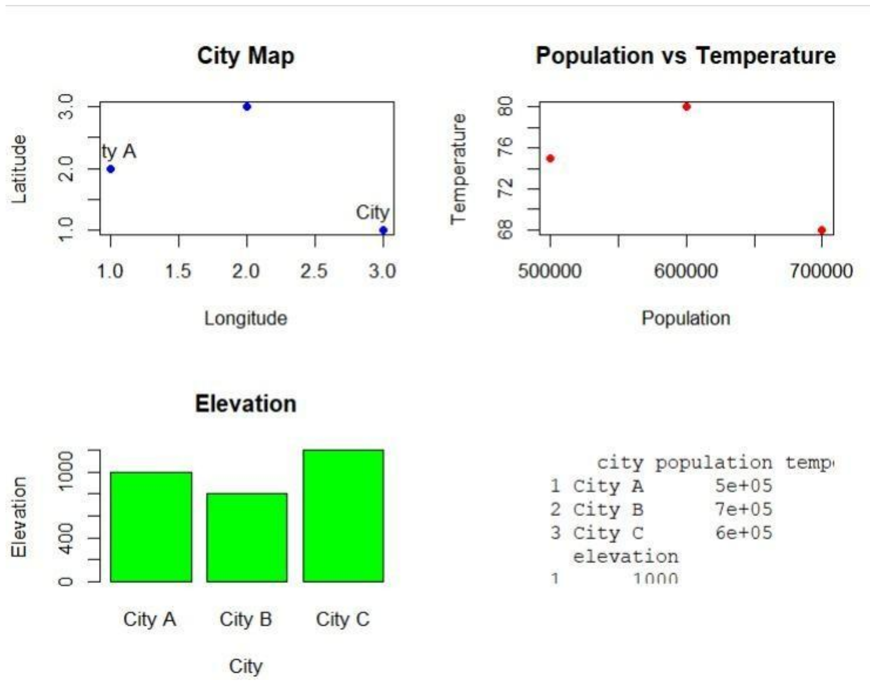
SET 13 – Q3 Display Geographic Data Table
geo <- data.frame(
 City=c("City A","City B","City C"),
 Population=c(500000,700000,600000),
 AvgTemperature=c(75,68,80),
 Elevation=c(1000,800,1200)
)

```
cat("Geographic Data\n") print(geo)
```

```
Geographic Data
> print(geo)
    City Population AvgTemperature Elevation
1 City A      5e+05             75      1000
2 City B      7e+05             68       800
3 City C      6e+05             80      1200
>
```

SET 13 – Q4
Dashboard Equivalent in R

```
install.packages(c("leaflet","ggplot2","gridExtra"))

library(leaflet) library(ggplot2)
library(gridExtra)

geo <- data.frame(
  City=c("City A","City B","City C"),
  Population=c(500000,700000,600000),
  AvgTemperature=c(75,68,80),
  Elevation=c(1000,800,1200),
  Latitude=c(13.0827,12.9716,17.3850),
  Longitude=c(80.2707,77.5946,78.4867)
)

# Table print(geo)

# Scatter Plot plot1 <- ggplot(geo,
aes(AvgTemperature,
Population,             label=City))+
geom_point(size=4,color="red")+
geom_text(vjust=-1)+
labs(title="Scatter Plot")

print(plot1)

# Map leaflet(geo) %>% addTiles()
%>% addMarkers(lng=~Longitude,
        lat=~Latitude,          popup=~City)
```

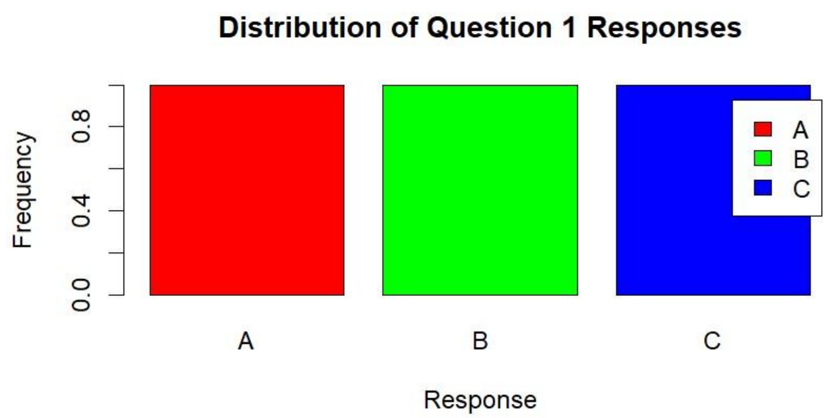


SET 14 – Q1

Stacked Bar Chart (Question 1 Responses) survey <-
data.frame(Respondent=c(1,2,3),
 Question1=c("A","B","C"), Question2=c("B","A","A"),
 Question3=c("C","D","B")
)

response_count <- table(survey$Question1)

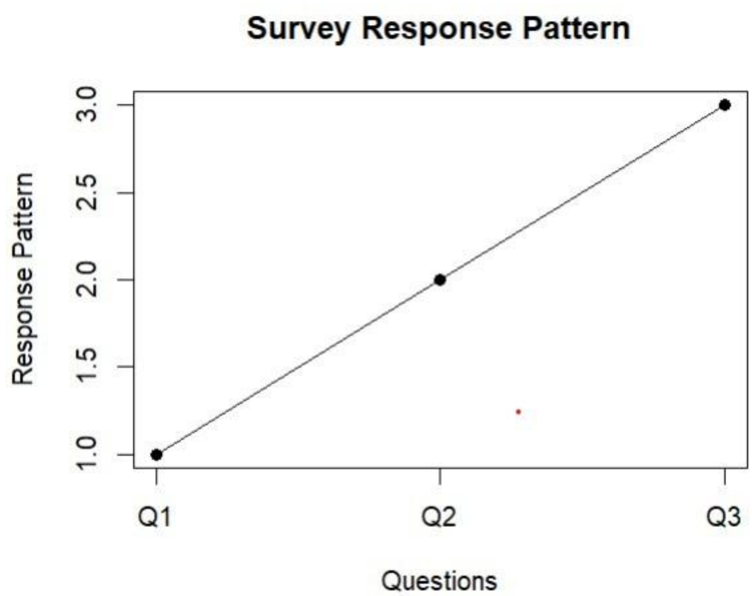
barplot(response_count, col=c("red","green","blue"),
main="Distribution of Question 1 Responses",
xlab="Response", ylab="Frequency",
 legend=rownames(response_count))



SET 14 – Q2

Generate a Radar Chart to visualize the overall pattern of responses.

```
# Install package (Run only once)
install.packages("fmsb") library(fmsb) #
Dataset survey <- data.frame(
  Respondent = c(1,2,3),
  Q1 = c(1,2,3),
  Q2 = c(2,1,1),
  Q3 = c(3,4,2)
)
# Radar Chart Data radar_data <-
rbind(   c(4,4,4),   # Maximum values
c(0,0,0),   # Minimum values
survey[1,2:4],   survey[2,2:4],
survey[3,2:4]
)
colnames(radar_data) <- c("Q1","Q2","Q3")
radarchart(   radar_data,   axistype=1,
pcol=c("red","blue","green"),   pfcol=c(rgb(1,0,0,0.3),
rgb(0,0,1,0.3),           rgb(0,1,0,0.3)),   plwd=2,
title="Radar Chart of Survey Responses"
)
legend(
  "topright",   legend=c("Respondent 1","Respondent 2","Respondent 3"),
col=c("red","blue","green"),   lty=1
)
```



SET 14 – Q3
Build a table to show survey response data.

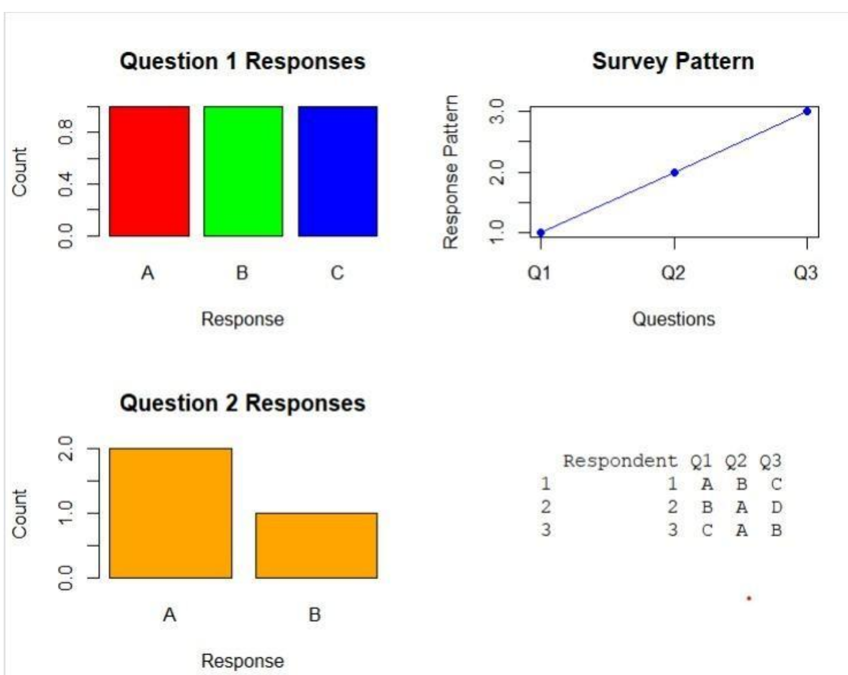
```
survey <- data.frame(   Respondent =
c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)

cat("Survey Results\n\n") print(survey)
```

```
>          print(survey)
  Respondent Q1 Q2 Q3
1          1  A  B  C
2          2  B  A  D
3          3  C  A  B
```

SET 14 – Q4
Dashboard (Equivalent in R)

```
library(ggplot2) library(fmsb)

survey <- data.frame(   Respondent =
c(1,2,3),
```

```
 Q1 = c("A","B","C"),
 Q2 = c("B","A","A"),
 Q3 = c("C","D","B")
)

# Table print(survey)

# Bar Chart
ggplot(survey,aes(Q1,fill=Q1))+
geom_bar()+ labs(title="Question 1
Responses",      x="Responses",
y="Count")

# Radar Chart radar
<- rbind( c(4,4,4),
c(0,0,0), c(1,2,3),
c(2,1,4), c(3,1,2)
)

colnames(radar)=c("Q1","Q2","Q3") radarchart(radar)

cat("Dashboard Components Displayed Successfully\n")
```



SET 15 – Q1

Create a Grouped Bar Chart install.packages("reshape2")
install.packages("ggplot2")

library(reshape2) library(ggplot2)

sales <- data.frame(
Product=c("Product A","Product B","Product C"),
January=c(2000,1500,1200),
February=c(2200,1800,1400),
March=c(2400,1600,1100)
)

sales_long <- melt(sales,id.vars="Product")

ggplot(sales_long,
aes(Product,value,fill=variable)
)+ geom_bar(
stat="identity", position="dodge")+
labs(title="Quarterly Product Sales",
x="Product", y="Sales"
)

SET 15 – Q2

Generate a Stacked Area Chart
library(reshape2) library(ggplot2)

```
sales <- data.frame(
Product=c("Product A","Product B","Product C"),
January=c(2000,1500,1200),
February=c(2200,1800,1400),
March=c(2400,1600,1100)
)

sales_long <- melt(sales,id.vars="Product")

ggplot( sales_long,
aes( x=variable,
y=value,
group=Product,
fill=Product
) )+ geom_area()+ labs(
title="Overall Sales Trend",
x="Month", y="Sales"
)
```

SET 15 – Q3
Build a table to show the monthly sales data for each product.

```
# Dataset sales <-
data.frame(
  ProductID = c(1,2,3),
```

```
  ProductName = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)

# Display Table cat("Monthly Product Sales
Data\n\n") print(sales)
```

```
> print(sales)
    Product   Jan  Feb  Mar
1 Product A 2000 2200 2400
2 Product B 1500 1800 1600
3 Product C 1200 1400 1100
```

SET 15 – Q4

Develop a dashboard combining the grouped bar chart, stacked area chart, and table.

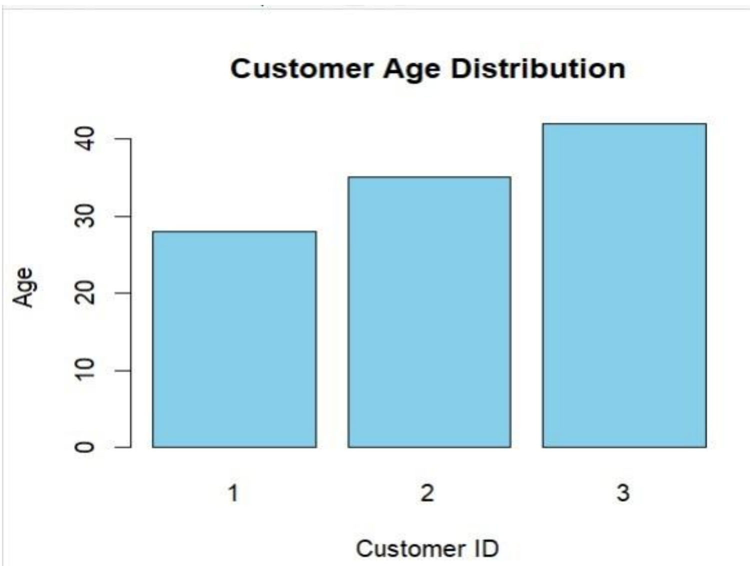
```
# Install packages (Run only once) install.packages("ggplot2")
install.packages("reshape2")

library(ggplot2) library(reshape2)

# Dataset sales <-
data.frame(
  Product = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)

# Display Table print(sales)

# Convert data to long format sales_long <- melt(sales,
id.vars="Product")

# Grouped Bar Chart ggplot(sales_long,       aes(Product,
value, fill=variable)) +   geom_bar(stat="identity",
position="dodge") +   labs(title="Grouped Bar Chart",
x="Product",       y="Sales")
```

```
# Stacked Area Chart ggplot(sales_long,
aes(variable, value,
group=Product,            fill=Product)) +
geom_area() +   labs(title="Stacked Area
Chart",       x="Month",       y="Sales")

cat("Dashboard Components Displayed Successfully")
```

SET 16 – Q1

Create a bar chart to visualize the distribution of customer ages.

```
# Install package (Run only once) install.packages("ggplot2")

library(ggplot2)

# Dataset customer <-
data.frame(   CustomerID =
c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

# Bar Chart ggplot(customer,
aes(x=factor(CustomerID),           y=Age,
fill=Gender)) +   geom_bar(stat="identity") +
labs(title="Customer Age Distribution",
x="Customer ID",       y="Age")
```



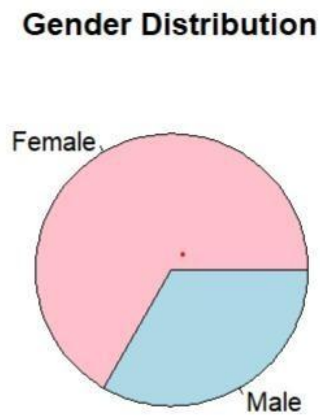
SET 16 – Q2

Generate a pie chart to represent the distribution of customers by gender. #
Dataset customer <- data.frame(CustomerID = c(1,2,3),
 Age = c(28,35,42),
 Gender = c("Female","Male","Female"),
 Income = c(50000,60000,75000)
)

# Count Gender gender_count <-
table(customer$Gender)

# Pie Chart pie(gender_count,
col=c("pink","lightblue"), main="Customer
Gender Distribution"
)

legend(
 "topright", legend=names(gender_count),
fill=c("pink","lightblue")
)



SET 16 – Q3

Build a table to show the demographic information for each customer.

```
# Dataset customer <-
data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

cat("Customer Demographics\n\n")

print(customer)
```

```
  Customer Age Gender Income
1        1  28 Female  50000
2        2  35   Male  60000
3        3  42 Female  75000
```

SET 16 – Q4

Develop a dashboard combining the bar chart, pie chart, and table.

```
# Install packages (Run only once) install.packages("ggplot2")

library(ggplot2)
```

```
# Dataset customer <-
data.frame(   CustomerID =
c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

# Display Table print(customer)

# Bar Chart ggplot(customer,
aes(x=factor(CustomerID),           y=Age,
fill=Gender)) +   geom_bar(stat="identity") +
labs(title="Customer Age Distribution",
x="Customer ID",        y="Age")

# Pie Chart gender_count <-
table(customer$Gender)

pie(gender_count,     col=c("pink","lightblue"),
main="Gender Distribution")

cat("Dashboard Components Displayed Successfully")
```

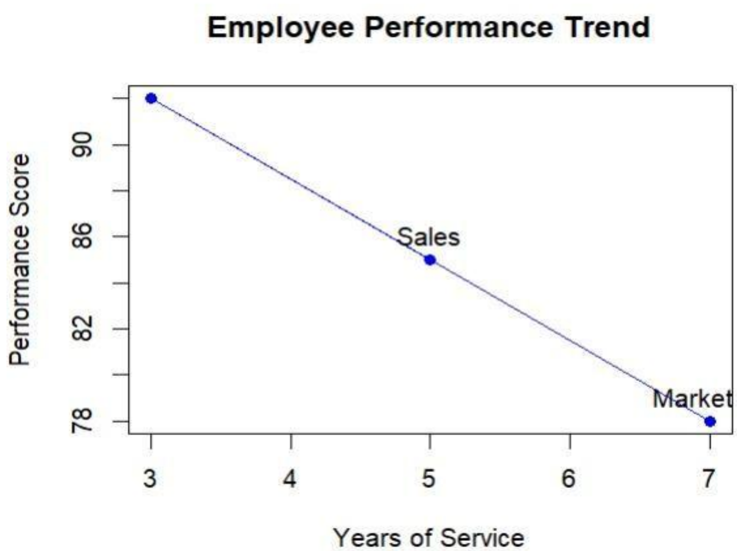
SET 17 – Q1

Create a line chart to visualize the performance trend of employees over time.

```
# Install package (Run only once) install.packages("ggplot2")

library(ggplot2)
```

Dataset employee <-
data.frame(EmployeeID =
c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)

Line Chart ggplot(employee,
aes(x=YearsService,   y=PerformanceScore,
group=1)) +   geom_line(color="blue", linewidth=1) +
geom_point(size=4, color="red") +
labs(title="Employee Performance Trend",
x="Years of Service",   y="Performance Score")



SET 17 – Q2 Generate a bar chart showing the distribution of employees across different departments.

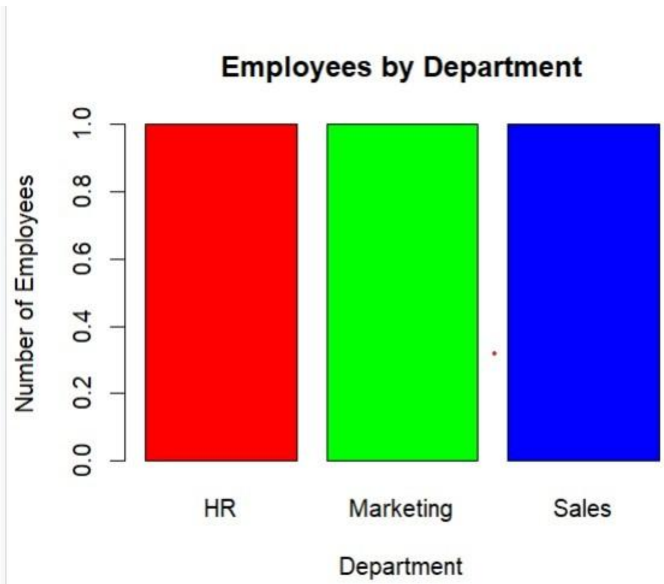
Install package (Run only once) install.packages("ggplot2")

library(ggplot2)

Dataset employee <-
data.frame(EmployeeID =
c(1,2,3),
  Department = c("Sales","HR","Marketing"),

YearsService = c(5,3,7),
PerformanceScore = c(85,92,78)
)

Bar Chart ggplot(employee,
aes(x=Department, fill=Department)) +
geom_bar() + labs(title="Employees by
Department", x="Department",
y="Number of Employees")



SET 17 – Q3

Build a table to display the performance data for each employee.

Dataset employee <-
data.frame(EmployeeID =
c(1,2,3),
Department = c("Sales","HR","Marketing"),
YearsService = c(5,3,7),
PerformanceScore = c(85,92,78)
)

cat("Employee Performance Data\n\n")

print(employee)

	Employee	Department	Years	Performance
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78

SET 17 – Q4

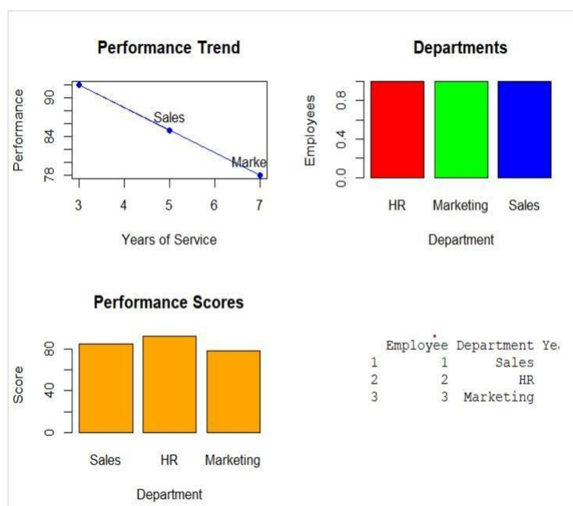
Develop a dashboard combining the line chart, bar chart, and table. # Install package (Run only once) install.packages("ggplot2")

library(ggplot2)

Dataset employee <-
data.frame(EmployeeID =
c(1,2,3),
 Department = c("Sales","HR","Marketing"),
 YearsService = c(5,3,7),
 PerformanceScore = c(85,92,78)
)

Display Table print(employee)

Line Chart ggplot(employee,
aes(x=YearsService, y=PerformanceScore,
group=1)) + geom_line(color="blue",
linewidth=1) + geom_point(size=4, color="red") +
labs(title="Performance Trend", x="Years of
Service", y="Performance Score")



Bar Chart ggplot(employee,
aes(x=Department,
fill=Department)) + geom_bar() +
labs(title="Employees by
Department", x="Department",
y="Count")

cat("Dashboard Components Displayed Successfully")